

Master Laplace Reference Guide

Prepared by Staples Education

Unit 12: The Laplace Transform — Foundations, Toolkit, Solving ODEs, Convolution, Advanced Applications, and Master Tables

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Unit 12 at a Glance: The Laplace Transform

The Laplace transform converts a differential equation in t into an algebraic equation in s . Initial data is baked in from the start, discontinuous and impulsive forcing become ordinary algebra, and the structure of a system shows up as poles in the complex plane. The entire unit is one pipeline: transform, solve the algebra, invert.

- **Foundations** — the defining integral, existence (exponential order), the Euler trick, and the $t = 0$ initial-condition traps.
- **The toolkit** — linearity, the two shift theorems, Heaviside and Dirac packaging, and the transforms of derivatives and integrals.
- **Solving ODEs** — the five-step method, partial fractions, discontinuous and impulse forcing, systems, and periodic inputs.
- **Convolution** — the convolution theorem, Duhamel's principle, Volterra integral equations, transfer functions, and stability.
- **Advanced** — PDEs by Laplace, Abel's integral equation, and resonance read off the s -plane.
- **Reference** — the master transform and property tables, the inversion playbook, and the initial/final-value (steady-state) theorems.

Part I — Foundations

12.1 The Definition and the Transform as an Operator

The Laplace transform takes a function $f(t)$ defined for $t \geq 0$ and produces a function $F(s)$ of a new variable s . It is an *integral operator*: it acts on a whole function and returns a whole function.

The Defining Integral

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

The integral runs over $[0, \infty)$, so the transform sees only the future ($t \geq 0$). The kernel e^{-st} is a decaying weight (for $\text{Re}(s) > 0$) that tames the growth of f and guarantees the integral converges in a half-plane.

The two base transforms everything else is built from come straight from the definition. For $f(t) = 1$:

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \right) = \frac{1}{s}, \quad s > 0.$$

For $f(t) = e^{at}$ the two exponentials merge:

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-(s-a)t} dt = \left[-\frac{1}{s-a} e^{-(s-a)t} \right]_0^\infty = \frac{1}{s-a}, \quad s > a.$$

Worked Example: $\mathcal{L}\{t\}$ from the Definition

The power rule $\mathcal{L}\{t^n\} = n!/s^{n+1}$ is itself a consequence of the definition. Integrate $\int_0^\infty t e^{-st} dt$ by parts with $u = t$, $dv = e^{-st} dt$:

$$\left[-\frac{t}{s} e^{-st} \right]_0^\infty + \frac{1}{s} \int_0^\infty e^{-st} dt = 0 + \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}.$$

The boundary term vanishes because e^{-st} kills the linear growth at infinity. Repeating the integration by parts n times produces the factorial and the s^{n+1} in the denominator.

Intuition: The Frequency Domain

Reading $F(s)$ as a function of s is reading f in the **frequency domain**: each value of s asks “how much of the mode e^{st} is inside f ?” Sharp features and oscillations in t become poles and resonances in s . The transform is a change of coordinates that makes the *structure* of a signal — its decay rates and frequencies — visible as the **location of poles** in the complex plane.

Key Takeaway

The transform is a machine that trades **calculus in t** for **algebra in s** . Differentiation becomes multiplication by s , integration becomes division by s , and that is the entire reason the method solves differential equations.

12.2 Existence: Exponential Order and Piecewise Continuity

Not every function has a transform. The integral must actually converge, and two mild conditions guarantee that.

Sufficient Conditions for Existence

If f is **piecewise continuous** on $[0, \infty)$ and of **exponential order** — meaning there exist constants $M > 0$, a , and T with

$$|f(t)| \leq M e^{at} \quad \text{for all } t \geq T,$$

then $F(s)$ exists for all $s > a$, and $F(s) \rightarrow 0$ as $s \rightarrow \infty$.

Common Pitfall: Functions That Grow Too Fast

$f(t) = e^{t^2}$ is *not* of exponential order — it outruns every $M e^{at}$ — so its Laplace transform does not exist. By contrast every polynomial, every e^{at} , every sin and cos, and any product of these is of exponential order. A quick existence test: if $\lim_{s \rightarrow \infty} F(s) \neq 0$, the candidate F cannot be a Laplace transform of a well-behaved f .

12.3 The Euler Trick: Sines, Cosines, and Complex Exponentials

Rather than integrate $e^{-st} \cos(\omega t)$ by parts twice, transform the complex exponential once and read off the real and imaginary parts.

Transforming Trig via $e^{i\omega t}$

1. Write $e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$.

2. Transform it as an ordinary exponential with $a = i\omega$:

$$\mathcal{L}\{e^{i\omega t}\} = \frac{1}{s - i\omega}.$$

3. Rationalize by multiplying by $\frac{s + i\omega}{s + i\omega}$:

$$\frac{s + i\omega}{s^2 + \omega^2} = \frac{s}{s^2 + \omega^2} + i \frac{\omega}{s^2 + \omega^2}.$$

4. Match real and imaginary parts:

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}.$$

The same trick delivers the hyperbolic pair (use $a = \pm\omega$ real) and, combined with the first shift theorem, the damped trig transforms in one motion.

12.4 The $t = 0$ Traps: One-Sided Limits and 0^-

Because the transform integrates from 0, the value of f and its derivatives *at the start* is special, and it is where careless bookkeeping breaks a solution.

Common Pitfall: Which Zero?

In $\mathcal{L}\{f'\} = sF(s) - f(0)$ the term $f(0)$ is the initial value *carried by the problem*, the one-sided limit $f(0^+)$ (or $f(0^-)$ in the convention that places impulses just before $t = 0$). When forcing contains a $\delta(t)$ at the origin, the 0^- convention is the safe one: it captures the impulse *inside* the integral instead of losing it at the lower limit. State your convention and use it consistently across every derivative term.

Key Takeaway

The transform method is an **initial-value** method by construction: the data $f(0), f'(0), \dots$ enters the algebra immediately through the derivative rules, so there is never a separate “apply the initial conditions” step at the end the way there is with undetermined coefficients.

Part II — The Core Toolkit

12.5 Linearity and the Building-Block Transforms

The transform is linear, which is what lets us break any input into a sum of known pieces.

Linearity

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}.$$

The same holds for the inverse: $\mathcal{L}^{-1}\{aF + bG\} = af + bg$.

The handful of transforms below, combined with linearity and the two shifts, covers the overwhelming majority of problems.

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad \mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}, \quad \mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}.$$

12.6 The First Shift Theorem (s -Shift): Multiplication by e^{at}

Multiplying $f(t)$ by e^{at} in the time domain *shifts the variable* in the s -domain.

First Shift Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a).$$

Equivalently, $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$: a shift of the argument becomes an exponential factor.

Worked Example: A Damped Cosine

Find $\mathcal{L}\{e^{-2t} \cos 3t\}$. Start from $\mathcal{L}\{\cos 3t\} = \frac{s}{s^2 + 9}$ and shift $s \mapsto s + 2$:

$$\mathcal{L}\{e^{-2t} \cos 3t\} = \left. \frac{s}{s^2 + 9} \right|_{s \rightarrow s+2} = \frac{s+2}{(s+2)^2 + 9}.$$

The denominator $(s+2)^2 + 9$ is exactly a completed square — the fingerprint of a damped oscillation hiding in any inverse transform.

12.7 The Second Shift Theorem (t -Shift): Heaviside and Piecewise Forcing

The unit step $u(t-c)$ (also written $u_c(t)$ or $H(t-c)$) is 0 before $t=c$ and 1 after. It is the switch that turns forcing on, and the second shift theorem is how a delayed, switched-on signal transforms.

Heaviside and the Second Shift

$$u(t-c) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}, \quad \mathcal{L}\{u(t-c)\} = \frac{e^{-cs}}{s}.$$

$$\boxed{\mathcal{L}\{u(t-c)f(t-c)\} = e^{-cs}F(s)} \quad \mathcal{L}^{-1}\{e^{-cs}F(s)\} = u(t-c)f(t-c).$$

A delay by c in time multiplies the transform by e^{-cs} .

Packaging a Piecewise Function for the Transform

1. Rewrite the piecewise $f(t)$ as a sum of steps: $f(t) = f_0(t) + \sum_k u(t - c_k) [f_{k+1}(t) - f_k(t)]$.
2. For each step, force the trailing factor into the shifted form $g(t - c_k)$ — i.e. **re-express every t as $(t - c_k) + c_k$** .
3. Transform term by term using $\mathcal{L}\{u(t - c)g(t - c)\} = e^{-cs}G(s)$.

Common Pitfall: Forgetting to Shift the Argument

$\mathcal{L}\{u(t - c)f(t)\}$ is **not** $e^{-cs}F(s)$. The theorem requires the *same* shift inside f . If the factor is $f(t) = t^2$ switched on at $c = 3$, write $t^2 = ((t - 3) + 3)^2 = (t - 3)^2 + 6(t - 3) + 9$ first, then transform each shifted piece. Skipping this step is the single most common Heaviside error.

Worked Example: Packaging a Two-Piece Function

Transform $f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 1, & t \geq 2 \end{cases}$. Write it as a base piece plus a switched correction at $c = 2$:

$$f(t) = t - u(t - 2)(t - 1).$$

The correction must be expressed in $(t - 2)$: since $t - 1 = (t - 2) + 1$,

$$f(t) = t - u(t - 2)[(t - 2) + 1].$$

Transform term by term with $\mathcal{L}\{t\} = 1/s^2$ and the second shift:

$$F(s) = \frac{1}{s^2} - e^{-2s} \left(\frac{1}{s^2} + \frac{1}{s} \right).$$

12.8 The Dirac Delta and Impulse Forcing

The Dirac delta $\delta(t - c)$ models an instantaneous unit impulse — a hammer blow, a spark, a sudden kick — concentrated entirely at $t = c$. It is not a function but a distribution, defined by how it acts inside an integral.

The Sifting Property and Its Transform

$$\int_{-\infty}^{\infty} \delta(t - c) g(t) dt = g(c) \quad (\text{sifting}), \quad \mathcal{L}\{\delta(t - c)\} = e^{-cs}, \quad \mathcal{L}\{\delta(t)\} = 1.$$

The delta is the derivative of the Heaviside step: $\delta(t - c) = \frac{d}{dt}u(t - c)$, which is why its transform is s times that of the step.

Key Takeaway

A δ -forcing produces a **jump in the highest-order state**: for $y'' + \dots = \delta(t - c)$ the velocity y' jumps by 1 at $t = c$ while y itself stays continuous. The response after the kick is exactly the system's free response restarted with that jolted initial condition — the **impulse response**.

Worked Example: Impulse-Forced Oscillator

Solve $y'' + y = \delta(t - \pi)$, $y(0) = 0$, $y'(0) = 0$. Transform with $\mathcal{L}\{\delta(t - \pi)\} = e^{-\pi s}$:

$$(s^2 + 1)Y = e^{-\pi s} \Rightarrow Y(s) = e^{-\pi s} \cdot \frac{1}{s^2 + 1}.$$

Since $\mathcal{L}^{-1}\{1/(s^2 + 1)\} = \sin t$, the second shift gives

$$y(t) = u(t - \pi) \sin(t - \pi) = -u(t - \pi) \sin t.$$

The mass is at rest until the kick at $t = \pi$, then oscillates freely — a clean picture of the impulse response switched on at the moment of the blow.

12.9 Transforms of Derivatives

This is the rule that makes the whole method work: differentiation in t becomes multiplication by s , minus the initial data.

Derivative Rules

$$\begin{aligned}\mathcal{L}\{f'\} &= sF(s) - f(0), \\ \mathcal{L}\{f''\} &= s^2F(s) - sf(0) - f'(0), \\ \mathcal{L}\{f^{(n)}\} &= s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0).\end{aligned}$$

The pattern: each derivative contributes one power of s and peels off one initial value, highest-order data attached to the highest power of s .

12.10 Transforms of Integrals; Multiplication and Division by t

The mirror image of the derivative rule, plus the two “polynomial weight” rules.

Integral and Weight Rules

$$\begin{aligned}\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} &= \frac{F(s)}{s}, & \mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} &= \int_0^t f(\tau) d\tau. \\ \mathcal{L}\{t^n f(t)\} &= (-1)^n \frac{d^n F}{ds^n}, & \mathcal{L}\left\{\frac{f(t)}{t}\right\} &= \int_s^\infty F(\sigma) d\sigma.\end{aligned}$$

Worked Example: Multiplication by t

Find $\mathcal{L}\{t \sin \omega t\}$. With $F(s) = \frac{\omega}{s^2 + \omega^2}$, apply $\mathcal{L}\{tf(t)\} = -F'(s)$:

$$-\frac{d}{ds} \left(\frac{\omega}{s^2 + \omega^2} \right) = -\omega \cdot \frac{-2s}{(s^2 + \omega^2)^2} = \frac{2\omega s}{(s^2 + \omega^2)^2}.$$

The squared denominator is the algebraic signature of *resonance* — a repeated complex pole — which we return to in Part V.

Part III — Solving ODEs

12.11 The Five-Step Transform Method

Every constant-coefficient initial value problem yields to the same loop.

The Five-Step Method

1. **Transform** both sides of the ODE, applying the derivative rules so the initial conditions enter immediately.
2. **Substitute** the given $y(0), y'(0), \dots$
3. **Solve algebraically** for $Y(s)$ — it is always a rational function (times e^{-cs} factors if the forcing is delayed).
4. **Decompose** $Y(s)$ by partial fractions / completing the square into table-ready pieces.
5. **Invert** term by term to recover $y(t)$.

Worked Example: A Full IVP

Solve $y'' - 3y' + 2y = 0$, $y(0) = 1$, $y'(0) = 0$. Transform and substitute:

$$(s^2Y - s) - 3(sY - 1) + 2Y = 0 \Rightarrow (s^2 - 3s + 2)Y = s - 3.$$

Solve and factor the denominator $s^2 - 3s + 2 = (s - 1)(s - 2)$:

$$Y(s) = \frac{s - 3}{(s - 1)(s - 2)} = \frac{2}{s - 1} - \frac{1}{s - 2}.$$

Invert each term:

$$y(t) = 2e^t - e^{2t}.$$

Worked Example: An Underdamped IVP, End to End

Solve $y'' + 2y' + 5y = 0$, $y(0) = 2$, $y'(0) = -2$. Transform with both derivative rules and substitute the data:

$$(s^2Y - 2s + 2) + 2(sY - 2) + 5Y = 0.$$

Collect Y and the constants:

$$(s^2 + 2s + 5)Y = 2s + 2 \Rightarrow Y(s) = \frac{2s + 2}{s^2 + 2s + 5}.$$

Complete the square in the denominator, $s^2 + 2s + 5 = (s + 1)^2 + 4$, and align the numerator to the shifted variable $s + 1$:

$$Y(s) = \frac{2(s + 1)}{(s + 1)^2 + 2^2}.$$

This is exactly 2 times a shifted cosine ($a = -1$, $\omega = 2$):

$$y(t) = 2e^{-t} \cos 2t.$$

The e^{-t} envelope (from the real part of the poles) and the $2t$ oscillation (from the imaginary part) are read straight off the completed square.

12.12 Inverse Transforms and Partial Fractions

Inversion is mostly the algebra of breaking a rational $Y(s) = P(s)/Q(s)$ into pieces the table recognizes.

Partial-Fraction Playbook

1. **Distinct real roots** $(s - r)$: term $\frac{A}{s - r} \rightarrow Ae^{rt}$.
2. **Repeated root** $(s - r)^m$: terms $\frac{A_1}{s - r} + \cdots + \frac{A_m}{(s - r)^m}$, with $\frac{A_k}{(s - r)^k} \rightarrow A_k \frac{t^{k-1}}{(k-1)!} e^{rt}$.
3. **Irreducible quadratic** $s^2 + bs + c$: **complete the square** to $(s + \alpha)^2 + \beta^2$, then match to shifted sin / cos via the first shift theorem.

Common Pitfall: Don't Factor a Complex-Root Quadratic

When Q has complex roots, resist factoring it into complex linear terms. Complete the square instead: $s^2 + 2s + 5 = (s + 1)^2 + 4$ inverts cleanly to e^{-t} times a sin / cos in $2t$. Factoring into $(s - (-1 + 2i))(s - (-1 - 2i))$ is correct but invites arithmetic mistakes a real reader will not catch.

Worked Example: A Repeated Root

Invert $Y(s) = \frac{s+3}{(s+1)^2}$. A repeated factor needs two terms, $\frac{A}{s+1} + \frac{B}{(s+1)^2}$. Clear denominators:

$$s+3 = A(s+1) + B \Rightarrow A=1, B=2.$$

Each term has a known inverse ($1/(s+1) \rightarrow e^{-t}$ and $1/(s+1)^2 \rightarrow te^{-t}$):

$$y(t) = e^{-t} + 2te^{-t} = (1+2t)e^{-t}.$$

The te^{-t} term — a polynomial times an exponential — is the fingerprint of a repeated real root, exactly as in the homogeneous solution from undetermined coefficients.

12.13 Discontinuous and Impulse Forcing in IVPs

The transform's signature advantage: a forcing term that switches on, off, or spikes is handled by the same five steps, with no patching of solutions across breakpoints.

Worked Example: Step-Forced System

Solve $y'' + y = u(t - \pi)$, $y(0) = 0$, $y'(0) = 0$. Transform, using $\mathcal{L}\{u(t - \pi)\} = e^{-\pi s}/s$:

$$(s^2 + 1)Y = \frac{e^{-\pi s}}{s} \Rightarrow Y(s) = e^{-\pi s} \cdot \frac{1}{s(s^2 + 1)}.$$

Decompose the s -free factor: $\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1}$, which inverts to $1 - \cos t$. Now apply the second shift ($t \mapsto t - \pi$, switched at π):

$$y(t) = u(t - \pi) [1 - \cos(t - \pi)] = u(t - \pi) [1 + \cos t].$$

The system sits still until $t = \pi$, then oscillates about its new equilibrium.

12.14 Systems of ODEs

For a coupled system, transform every equation, collect the unknowns $X(s), Y(s), \dots$ into a linear algebraic system, and solve it (Cramer's rule or elimination) before inverting.

Transforming a System

1. Transform each equation; the derivative rules pull in each variable's initial data.
2. Write the result as $A(s)\mathbf{X}(s) = \mathbf{b}(s)$, a matrix of rational functions.
3. Solve for each $X_i(s)$, then invert componentwise.

The poles of the solution are the roots of $\det A(s)$ — the system's characteristic frequencies.

Worked Example: A Coupled First-Order System

Solve $x' = y$, $y' = -x$ with $x(0) = 1$, $y(0) = 0$. Transform both equations:

$$sX - 1 = Y, \quad sY - 0 = -X.$$

Substitute the first into the second, $s(sX - 1) = -X$, and solve for X :

$$(s^2 + 1)X = s \Rightarrow X(s) = \frac{s}{s^2 + 1}, \quad Y(s) = sX - 1 = \frac{-1}{s^2 + 1}.$$

Inverting each:

$$x(t) = \cos t, \quad y(t) = -\sin t.$$

The pair traces a circle — the phase portrait of a pure rotation, with poles exactly on the imaginary axis at $s = \pm i$.

12.15 Periodic Forcing Functions

A forcing that repeats with period T has a transform expressible over a single period, summed by a geometric series.

Transform of a T -Periodic Function

If $f(t + T) = f(t)$ for $t \geq 0$, then

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

The factor $\frac{1}{1 - e^{-sT}} = \sum_{n \geq 0} e^{-nsT}$ is exactly the second-shift bookkeeping for the infinitely repeated copies of one period.

Worked Example: Square Wave

For the square wave that is 1 on $[0, a)$ and 0 on $[a, 2a)$ with period $T = 2a$, only the first half-period contributes to the integral:

$$\int_0^{2a} e^{-st} f(t) dt = \int_0^a e^{-st} dt = \frac{1 - e^{-as}}{s}.$$

$$\mathcal{L}\{f\} = \frac{1 - e^{-as}}{s(1 - e^{-2as})} = \frac{1}{s(1 + e^{-as})},$$

using $1 - e^{-2as} = (1 - e^{-as})(1 + e^{-as})$.

Part IV — Convolution and Systems Analysis

12.16 The Convolution Theorem

A product in the s -domain corresponds not to a product in t but to a *convolution* — a blending integral.

Convolution Theorem

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau, \quad \mathcal{L}\{f * g\} = F(s) G(s).$$

Convolution is commutative ($f * g = g * f$), associative, and distributes over addition.

Common Pitfall: $\mathcal{L}\{fg\} \neq F(s)G(s)$

The transform of a *product* is not the product of transforms. $F(s)G(s)$ inverts to the *convolution* $f * g$, never to $f(t)g(t)$. Use convolution to invert a transform that is naturally a product — especially $Y(s) = H(s) \cdot R(s)$, system times input.

Worked Example: Inverting a Product by Convolution

Invert $Y(s) = \frac{1}{s^2(s^2 + 1)}$ by reading it as $\frac{1}{s^2} \cdot \frac{1}{s^2 + 1}$, i.e. $f(t) = t$ convolved with $g(t) = \sin t$:

$$y(t) = \int_0^t \tau \sin(t - \tau) d\tau.$$

Integrate by parts (or use the angle expansion):

$$y(t) = [\tau \cos(t - \tau)]_0^t - \int_0^t \cos(t - \tau) d\tau = t - \sin t.$$

A check confirms it: partial fractions give $\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1}$, which also inverts to $t - \sin t$.

12.17 Duhamel's Principle and the Impulse Response

The impulse response $h(t) = \mathcal{L}^{-1}\{H(s)\}$ is the system's reaction to a unit δ -kick. Duhamel's principle says the response to *any* input is the input convolved with that single fingerprint.

Duhamel / Zero-State Response

For a system with transfer function $H(s)$ and input $r(t)$ (zero initial state),

$$Y(s) = H(s)R(s) \iff y(t) = (h * r)(t) = \int_0^t h(t - \tau) r(\tau) d\tau.$$

Key Takeaway

Know the **impulse response** and you know the response to everything: steps, ramps, sinusoids, arbitrary signals — all obtained by one convolution. This is why $h(t)$ (equivalently $H(s)$) is treated as the complete characterization of a linear time-invariant system.

12.18 Volterra Integral Equations

An equation with the unknown inside a convolution integral becomes algebra under the transform.

Worked Example: A Convolution-Type Volterra Equation

Solve $y(t) = t + \int_0^t y(\tau) \sin(t - \tau) d\tau$. The integral is $y * \sin t$, so transform both sides with $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$:

$$Y = \frac{1}{s^2} + Y \cdot \frac{1}{s^2+1} \Rightarrow Y \left(1 - \frac{1}{s^2+1}\right) = \frac{1}{s^2}.$$

Simplify $1 - \frac{1}{s^2+1} = \frac{s^2}{s^2+1}$:

$$Y \cdot \frac{s^2}{s^2+1} = \frac{1}{s^2} \Rightarrow Y(s) = \frac{s^2+1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}.$$

$$y(t) = t + \frac{t^3}{6}.$$

12.19 Transfer Functions

For a linear time-invariant system $a_n y^{(n)} + \dots + a_0 y = r(t)$ at rest, the transfer function is the ratio of output to input transform.

Transfer Function

$$H(s) = \frac{Y(s)}{R(s)} = \frac{1}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0} = \frac{1}{p(s)},$$

where $p(s)$ is the system's characteristic polynomial. The numerator and denominator encode, respectively, the system's zeros and poles.

Worked Example: Step Response of a First-Order System

Take $\tau y' + y = r(t)$ at rest, so $H(s) = \frac{1}{\tau s + 1}$. The unit-step input $r(t) = u(t)$ has $R(s) = 1/s$, so

$$Y(s) = H(s)R(s) = \frac{1}{s(\tau s + 1)} = \frac{1}{s} - \frac{\tau}{\tau s + 1}.$$

Inverting (the second term is $\frac{1}{s+1/\tau}$ in disguise):

$$y(t) = 1 - e^{-t/\tau}.$$

The output relaxes to its steady state 1 on the time scale τ — the classic exponential-approach step response of a first-order lag.

12.20 Stability, Poles, and the Pole Diagram

The poles of $H(s)$ — the roots of $p(s)$ — are the natural frequencies, and their location in the complex plane dictates every qualitative behavior.

Poles and Behavior

A pole at $s = \alpha + i\beta$ contributes a mode $e^{\alpha t} \cos(\beta t)$ (or sin). Therefore:

$$\begin{aligned}\alpha < 0 \text{ (left half-plane)} &\Rightarrow \text{decaying mode (stable),} \\ \alpha = 0 \text{ (imaginary axis)} &\Rightarrow \text{sustained oscillation (marginal),} \\ \alpha > 0 \text{ (right half-plane)} &\Rightarrow \text{growing mode (unstable).}\end{aligned}$$

β sets the oscillation frequency; α sets the growth or decay rate.

Key Takeaway

A system is **BIBO stable** exactly when *all* poles lie strictly in the left half-plane. A single repeated pole on the imaginary axis is the algebraic face of **resonance**: a bounded input produces an unbounded, t -growing output.

Part V — Advanced Applications

12.21 The Fourier Connection

The Laplace transform is a generalized Fourier transform: it adds a real decay factor $e^{-\sigma t}$ so that functions which never settle (and so have no Fourier transform) still possess a Laplace transform.

Laplace as a Damped Fourier Transform

Write $s = \sigma + i\omega$. Then

$$F(s) = \int_0^\infty e^{-(\sigma+i\omega)t} f(t) dt = \int_0^\infty [e^{-\sigma t} f(t)] e^{-i\omega t} dt.$$

This is the Fourier transform of the *damped, one-sided* signal $e^{-\sigma t} f(t)$. Setting $\sigma = 0$ (evaluating on the imaginary axis $s = i\omega$) recovers the Fourier transform itself, when it exists.

Key Takeaway

The imaginary axis $s = i\omega$ is the **Fourier axis**: the system's frequency response is just $H(s)$ evaluated there, $H(i\omega)$. The real part σ of s is the convergence knob Fourier lacks — it is what lets Laplace handle growing and switched-on transients that pure Fourier analysis cannot.

12.22 PDEs via the Laplace Transform

Transforming in t turns a PDE in (x, t) into an *ordinary* differential equation in x , with s riding along as a parameter.

Laplace Method for an Evolution PDE

1. Apply $\mathcal{L}_t$ to the PDE. Time derivatives become algebra ($u_t \rightarrow sU - u(x, 0)$); spatial derivatives survive as $U_{xx} = \partial_{xx}U(x, s)$.
2. Solve the resulting ODE in x for $U(x, s)$, applying the boundary conditions (also transformed) and discarding solutions that blow up.
3. Invert $U(x, s)$ in s to recover $u(x, t)$.

Worked Example: Heat Equation on a Half-Line

For $u_t = k u_{xx}$, $x > 0$, with $u(x, 0) = 0$, $u(0, t) = u_0$, and u bounded as $x \rightarrow \infty$, transforming gives

$$sU = kU_{xx} \Rightarrow U_{xx} - \frac{s}{k}U = 0.$$

The bounded solution is $U(x, s) = A(s)e^{-\sqrt{s/k}x}$, and $U(0, s) = u_0/s$ fixes $A = u_0/s$:

$$U(x, s) = \frac{u_0}{s} e^{-\sqrt{s/k}x} \Rightarrow u(x, t) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right).$$

The $e^{-\sqrt{s}x}$ signature is the hallmark of diffusion and inverts to the complementary error function.

12.23 Abel's Integral Equation

Abel's equation — the tautochrone problem of finding a curve of constant descent time — is a convolution against a half-power kernel, tailor-made for the transform.

Abel's Equation and Its Kernel

$$\int_0^t \frac{\phi(\tau)}{\sqrt{t-\tau}} d\tau = f(t), \quad \mathcal{L}\{t^{-1/2}\} = \sqrt{\frac{\pi}{s}}.$$

The left side is $\phi * t^{-1/2}$, so transforming gives $\Phi(s)\sqrt{\pi/s} = F(s)$, hence $\Phi(s) = \sqrt{s/\pi} F(s)$, which inverts to a $\frac{1}{2}$ -order derivative of f .

Key Takeaway

Half-integer powers of t transform to half-integer powers of s through $\mathcal{L}\{t^a\} = \frac{\Gamma(a+1)}{s^{a+1}}$. This is what lets the Laplace method dispatch the **fractional-order** kernels that appear in Abel-type and diffusion problems.

12.24 Resonance Analysis in the s -Domain

Resonance is the case where the forcing frequency coincides with a natural frequency — algebraically, when forcing places a pole on top of an existing system pole.

Worked Example: Pure Resonance

Solve $y'' + \omega^2 y = \cos \omega t$, $y(0) = y'(0) = 0$. Transforming:

$$(s^2 + \omega^2)Y = \frac{s}{s^2 + \omega^2} \Rightarrow Y(s) = \frac{s}{(s^2 + \omega^2)^2}.$$

The *repeated* pole at $s = \pm i\omega$ is the signature of resonance. Inverting (via $\mathcal{L}\{t \sin \omega t\}$ from Part II):

$$y(t) = \frac{t}{2\omega} \sin \omega t.$$

The factor of t — a growing envelope — is exactly what the doubled pole predicts.

Worked Example: Beats (Near Resonance)

Drive off-frequency: $y'' + \omega^2 y = \cos \nu t$ with $\nu \neq \omega$, at rest. Transforming,

$$Y(s) = \frac{s}{(s^2 + \omega^2)(s^2 + \nu^2)}.$$

Now the poles at $\pm i\omega$ and $\pm i\nu$ are *distinct*, so partial fractions give a difference of two bounded cosines:

$$y(t) = \frac{\cos \nu t - \cos \omega t}{\omega^2 - \nu^2}.$$

As $\nu \rightarrow \omega$ the two frequencies nearly cancel, producing the slow amplitude-modulated envelope of **beats** — and in the exact limit the denominator $\omega^2 - \nu^2 \rightarrow 0$ forces the repeated pole and the t -growth of pure resonance.

Common Pitfall: Resonance Is a Repeated Pole, Not a Big One

Resonance does not come from a pole being “large”; it comes from a pole being *repeated*. A simple pole on the imaginary axis gives bounded oscillation; the **double** pole there is what produces the t -growth. Always check pole *multiplicity*, not just location, before declaring a system resonant.

Part VI — Master Reference

12.25 Master Transform Table

Core Transform Pairs

$f(t)$	$F(s)$	$f(t)$	$F(s)$
1	$\frac{1}{s}$	e^{at}	$\frac{1}{s-a}$
t^n	$\frac{n!}{s^{n+1}}$	$t^a \ (a > -1)$	$\frac{\Gamma(a+1)}{s^{a+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
$\sinh \omega t$	$\frac{\omega}{s^2 - \omega^2}$	$\cosh \omega t$	$\frac{s}{s^2 - \omega^2}$
$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2 + \omega^2}$	$e^{at} \cos \omega t$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$t \sin \omega t$	$\frac{2\omega s}{(s^2 + \omega^2)^2}$	$t \cos \omega t$	$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
$u(t-c)$	$\frac{e^{-cs}}{s}$	$\delta(t-c)$	e^{-cs}

12.26 Master Properties / Operational Table

Operational Rules

Linearity	$\mathcal{L}\{af + bg\} = aF + bG$
First shift (s)	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Second shift (t)	$\mathcal{L}\{u(t-c)f(t-c)\} = e^{-cs}F(s)$
Derivative	$\mathcal{L}\{f^{(n)}\} = s^n F - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$
Integral	$\mathcal{L}\left\{\int_0^t f\right\} = F(s)/s$
Multiply by t^n	$\mathcal{L}\{t^n f\} = (-1)^n F^{(n)}(s)$
Divide by t	$\mathcal{L}\{f/t\} = \int_s^\infty F(\sigma) d\sigma$
Convolution	$\mathcal{L}\{f * g\} = F(s)G(s)$
Periodic (T)	$\mathcal{L}\{f\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f dt$

12.27 Inversion Techniques Summary

Choosing an Inversion Route

1. **Table match** — recognize the pair directly, after completing the square or factoring out an e^{-cs} .
2. **Partial fractions** — for any proper rational $Y(s) = P/Q$; the workhorse of Part III.
3. **First/second shift** — peel off e^{at} factors (s -shift) and delays (e^{-cs} , t -shift).
4. **Convolution** — when $Y = F \cdot G$ is a product with no clean partial-fraction split (Part IV).
5. **Derivative-of-transform** — when Y looks like $-G'(s)$, read it as $t g(t)$.

12.28 Completing the Square: The Inversion Patterns

Almost every irreducible-quadratic inversion reduces to one of four shifted-trig patterns. Completing the square turns the denominator into $(s + a)^2 + \omega^2$; the numerator then decides which pattern applies.

The Four Shifted-Trig Inverses

$F(s)$	$f(t)$
$\frac{\omega}{(s + a)^2 + \omega^2}$	$e^{-at} \sin \omega t$
$\frac{s + a}{(s + a)^2 + \omega^2}$	$e^{-at} \cos \omega t$
$\frac{1}{(s + a)^2 + \omega^2}$	$\frac{1}{\omega} e^{-at} \sin \omega t$
$\frac{s}{(s + a)^2 + \omega^2}$	$e^{-at} \left(\cos \omega t - \frac{a}{\omega} \sin \omega t \right)$

Matching a Numerator to the Pattern

1. Complete the square: $s^2 + bs + c \rightarrow (s + a)^2 + \omega^2$ with $a = b/2$ and $\omega = \sqrt{c - a^2}$.
2. Rewrite the numerator in terms of $(s + a)$: replace s with $(s + a) - a$.
3. The $(s + a)$ part becomes a cos term; the leftover constant becomes a sin term (scaled by $1/\omega$).

Key Takeaway

A completed square $(s + a)^2 + \omega^2$ is the pole pair $s = -a \pm i\omega$ written algebraically. Reading a and ω off the denominator hands you the **decay rate** and the **oscillation frequency** of the answer before you finish the inversion — the same information the pole diagram shows geometrically.

12.29 Initial- and Final-Value Theorems (Steady-State Analysis)

These two theorems read off the endpoints of $f(t)$ *directly from* $F(s)$, without inverting — the fast route to steady-state and transient behavior.

Initial- and Final-Value Theorems

$$f(0^+) = \lim_{s \rightarrow \infty} s F(s), \quad \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s).$$

Worked Example: Steady State Without Inverting

Reuse the step response $Y(s) = \frac{1}{s(\tau s + 1)}$. The final-value theorem reads the long-run value straight off:

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s(\tau s + 1)} = \lim_{s \rightarrow 0} \frac{1}{\tau s + 1} = 1,$$

matching $1 - e^{-t/\tau} \rightarrow 1$ without ever inverting. The initial value checks too: $\lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} \frac{1}{\tau s + 1} = 0 = y(0)$.

Common Pitfall: The Final-Value Theorem Has Fine Print

The final-value theorem is valid **only when** $\lim_{t \rightarrow \infty} f(t)$ actually exists — equivalently, when every pole of $sF(s)$ lies strictly in the left half-plane. Applied blindly to $F(s) = \frac{\omega}{s^2 + \omega^2}$ it would “predict” 0, yet $\sin \omega t$ has no limit. Always confirm the poles before trusting the steady-state value.

Master Key Takeaway

The Laplace transform is one disciplined pipeline: **transform** $\rightarrow$ **solve the algebra** $\rightarrow$ **invert**. Initial data rides in through the derivative rules, discontinuities and impulses become e^{-cs} and δ bookkeeping, products become convolutions, and the geometry of the poles in the s -plane tells you — before you invert a single term — whether the system decays, oscillates, or blows up.